

Anik Ghosh

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PROFILE

I have 2.5+ years of experience as a machine learning engineer with specialisation in building and deploying production-grade **ML** and **Generative AI** systems, with a strong track record in building transformer-based NLP systems, RAG pipelines, and production MLOps infrastructure. Seeking roles in ML Engineering or GenAI where I can drive measurable business impact.

WORK EXPERIENCE

Machine Learning and Data Engineer | Valuechain Technology Ltd

April 2024 to Feb 2025

- Automated data extraction using Selenium and built an ETL pipeline using Apache Airflow, ensuring efficient data ingestion and transformation. The data was stored using Amazon Redshift.
- Designed and implemented a recommendation engine using transformer models that improved tender matching accuracy by 25% and deployed it using Docker for scalable production deployment.
- Built and deployed a production-ready web application using Django, with AWS S3 for file storage, Tailwind CSS for the front-end, and hosted on Heroku for rapid deployment.
- Worked with cross-functional teams and analysed user behaviour using Mixpanel to identify drop-off points and engagement trends, leading to data-driven product enhancements and improve user retention rates by 15% and created data visualisations using PowerBI.
- Built an LLM-powered email triage system using fine-tuned DistilBERT model and Gmail API to classify intents across 10 categories, achieving 97% classification accuracy and automating downstream workflows that previously required manual review.
- Used MLflow and Prometheus to deploy and monitor ML models in production. Applied MLOps practices like CI/CD (GitHub Actions), DVC-based version control, and automated retraining to improve accuracy and ensure model reliability over time.

Associate Software Engineer | Victoria's Secret & Co

Nov 2021 to Sep 2022

- Redesigned and enhanced an internal stock management tool using Node.js, Express, and Oracle SQL, improving database integration and overall user experience, resulting in smoother inventory tracking and faster data updates.
- Developed data pipelines using Python which would gather shipment data and generate detailed reports which reduced manual effort by 56%. The data was stored in MS SQL.
- Developed and automated several test cases using Selenium and Behaviour Driven Development (BDD) using Cucumber which reduced the overall functional test completion time by 30%.
- Configured CI/CD pipelines to run Selenium tests automatically using Jenkins/GitHub Actions, reducing manual effort and improving test reliability.

Programmer Analyst | Cognizant Technology Solutions

Feb 2021 to May 2021

- Successfully completed training on web technologies such as HTML, CSS, Javascript, and Azure Cloud Platform and received outstanding feedback on my final presentation and project.
- Built a client issue tracking system (C#, ASP.NET MVC) and optimised SQL queries for data processing pipelines during the 3-month analyst program at Cognizant.

Front End Web Developer | Ericsson India

May 2020 to Jul 2020

- Developed a single page web application for maintaining a list of company employees and their profile.
- Built an employee directory SPA using Angular, HTML, and CSS. Fundamental rules of UI/UX were implemented to make the website readable, navigable and accessible.

EDUCATION

University of Liverpool - Master of Science Data Science and AI

Sept 2022 - Sept 2023

- Dissertation: Human Activity recognition under limited labelled data using few-shot and transfer learning on wearable sensors.
- Key Modules: Deep Learning, NLP, Computer Vision, Statistical ML; Graduated with **Distinction**

Vellore Institute of Technology - Bachelor of Technology Electronics and Communications Engineering

July 2017 - June 2021

- Graduated with an overall **CGPA of 8.59/10**

SKILLS

- C++, Java, Python (Pytorch, Pandas, Tensorflow, Scikit Learn, Langchain, Matplotlib), C#
- Automation Testing, Manual Testing, Selenium, Unit Testing, UI Testing, A/B Testing
- HTML, CSS, React, Node.js , TypeScript, Javascript, RESTful API, FastAPI, Django
- MYSQL, Oracle SQL, SQLite, MongoDB, SQL Server, Database Design
- Data Visualisation (Matplotlib and Power BI), Data Mining, Data Modelling , Machine Learning (supervised and unsupervised learning) , Statistical Analysis, Deep Learning, Natural Language Processing and Large Language Models
- GIT, Apache Airflow, Docker, Apache Spark, Kubernetes, Amazon Web Service, Azure, Linux, CI/CD

PROJECTS

Human Activity Recognition using Limited Training Data (MSc Dissertation)

- Built a wearable sensor activity classifier (PAMAP2 dataset, 18 activity classes) comparing zero-shot, few-shot, and transfer learning approaches, targeting real-world scenarios where labelled training data is scarce, such as healthcare monitoring or fitness tracking apps.
- Compared zero-shot, few-shot, and transfer learning approaches on PAMAP2 across subject-disjoint splits; transfer learning achieved ~75% macro F1 vs ~30% zero-shot baseline across 18 activity classes, with rigorous collapse diagnostics and confusion matrix analysis.
- Conducted rigorous, leakage-free evaluation using subject-disjoint splits, confusion matrices, and collapse diagnostics, highlighting real-world deployment challenges in wearable sensor ML.

Job Recommendation Engine using GenAI

Nov 2025 - present

- Designed a NLP pipeline using Retrieval-Augmented Generation (RAG) to recommend relevant jobs from a large corpus of job descriptions based on semantic similarity with a candidate resume.
- Generated vector embeddings for resumes and job descriptions using **Sentence-BERT**, and applied similarity scoring and ranking to identify best-matching roles.
- Integrated an LLM via API to generate structured natural language summaries explaining match strengths and gaps between a resume and each recommended job.
- System returned relevant roles for 90% of test resumes and LLM-generated match summaries reduced time-to-shortlist by eliminating manual job description review.

YouTube Trending Video Analysis & Ad Performance Prediction

July 2025 - October 2025

- Utilised a Kaggle dataset to analyse trending YouTube videos and identify the types of videos best suited for successful ad campaigns.
- Built an end-to-end data pipeline on AWS using Amazon S3, AWS Glue, and Athena to ingest, transform, and analyse a Kaggle dataset of trending YouTube videos.
- Developed machine learning models (Logistic Regression, Random Forest, XGBoost) to predict ad suitability and engagement potential, using features derived from video metadata and early performance signals.
- Created an interactive Streamlit dashboard to visualise trending patterns, model predictions, and feature importance, and deployed the application on AWS EC2 using Docker.

Flight Booking Website

March 2025 - June 2025

- Built a full-stack flight booking web app using Node.js, Express.js, MongoDB, and Bootstrap to simulate an airline reservation system.
- Implemented features for searching flights, and performing CRUD operations on bookings (create, view, update, cancel).
- Designed a RESTful API backend with modular routing, middleware, and Mongoose-based data models for maintainability.
- Followed best practices in code organisation, error handling, and basic input validation to ensure a clean, scalable architecture.

EMNIST dataset image classification

Jan 2023 - March 2023

- Developed and trained MLP and CNN model using Python to classify the given handwritten characters to their respective labels.
- The model's performance was analysed and improved through hyper-parameter tweaking.

ACHIEVEMENTS

- Led a team to 11th place (top 3.4%) out of 327 teams in a smart building design hackathon. Proposed and presented an IoT + ML-based energy optimisation solution.

